DAY 2 Getting the Data

Project: Predicting Physical and Electronic Properties of Crystalline Solids from their Text Descriptions

Architecture block done today: 1. Data Acquisition

Date: Thursday, 22 Oct 2026

GOAL OF DAY 2

Download the TextEdge dataset, look inside it, and save what we learned to GitHub.

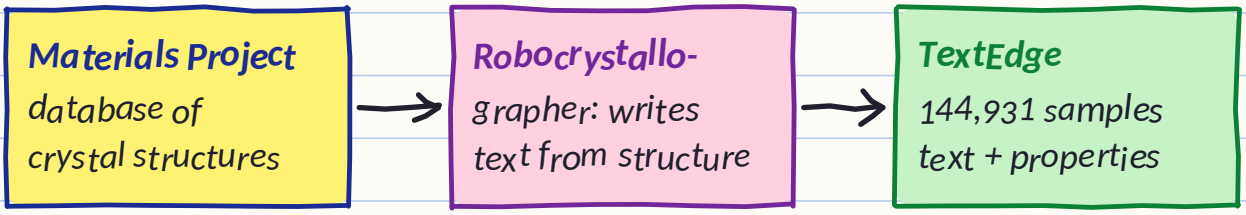
What we finished today:

- ✓ Found the authors' GitHub page (vertaix/LLM-Prop)
- ✓ Downloaded TextEdge (about 700 MB zip) from Google Drive
- ✓ Moved it into data/ and extracted it
- ✓ Found the exact paths of train, validation, test CSV files
- ✓ Fixed the libraries that were missing in my venv
- ✓ Looked at the 7 columns and picked input + outputs
- ✓ Wrote explore_data.py and ran it on the full data
- ✓ Saved the findings in notes.md and pushed to GitHub

NOTE: Day 1 + Day 2 done = 2 of 20 days. Block 1 (Data Acquisition) of the architecture is finished. Next is Block 2 (Text Preparation).

1. Where does the data come from?

Block 1 of the architecture



✓ The authors already did all of this and shared the finished dataset. So I do NOT need to rebuild it, I only download it.

19 Open the authors' code page

- The paper says (footnote 1): 'Code and data can be found at github.com/vertaix/LLM-Prop'.
- Folders I saw: data/samples, scripts, stopwords/en, tokenizers, figures, checkpoints/samples

x data/samples only has SMALL test samples, not the real dataset.

20 Make a data folder and download the full dataset

```
> mkdir data
```

(.gitignore already has data/ so it will never go to GitHub)

- The README links the full TextEdge on Google Drive, about 700 MB. I opened the link and the zip downloaded.
- If Google says it can't scan for viruses, click Download anyway.

2. Unzip and find the files

steps 21 to 23

21 Move the zip into LLM_PROP\data

Downloads folder > right-click the zip > Cut. Then open LLM_PROP\data > right-click > Paste.

- Why: everything for the project stays in one place and Downloads stays clean.

22 Extract it there

Right-click the zip > Extract All... > Extract. The big files take a minute.

23 Ask the terminal for the exact file paths

```
> dir data\*.csv /s /b
```

output: `...\data\TextEdge_v.1\TextEdge_v.1\train.csv  (and 5 more)`

My 3 real files (same folder):

`train.csv`	about 569 MB	(the big one, for learning)
`validation.csv`	about 45 MB	(to tune and pick the best model)
`test.csv`	about 51 MB	(final exam, used only at the end)

NOTE: Ignore __MACOSX and every file starting with ._ (1 KB each). They are junk left by a Mac when the zip was made.

WATCH OUT: NEVER double-click train.csv. 569 MB will freeze Excel and Excel can damage big CSV files. We read it safely with Python (pandas).

3. Peek inside train.csv

steps 24 and 25 (and one error I fixed)

24 Read only the first 3 rows

```
> python -c "import pandas as pd; df = pd.read_csv(
>     'data/TextEdge_v.1/TextEdge_v.1/train.csv', nrows=3);
>     print(df.columns.tolist()); print(df.iloc[0]...)"
```

(typed as ONE single line in the terminal, shown on 3 lines here to fit the page)

- nrows=3 means only 3 rows are read, so the 569 MB file is not fully loaded.

ERROR I got:

output: `ModuleNotFoundError: No module named 'pandas'`

- Meaning: my libraries were not inside the venv (I installed them when (venv) was not on).

FIX:

```
> pip install pandas numpy matplotlib scikit-learn tqdm
> python -c "import pandas, numpy, sklearn, torch, transformers; print('all good')"
```

output: `all good`

HABIT: (venv) switches OFF every time I close the terminal. Every new terminal: open in LLM_PROP, run venv\Scripts\activate and check that (venv) shows.

25 Check Git ignores the big data

```
> git status
```

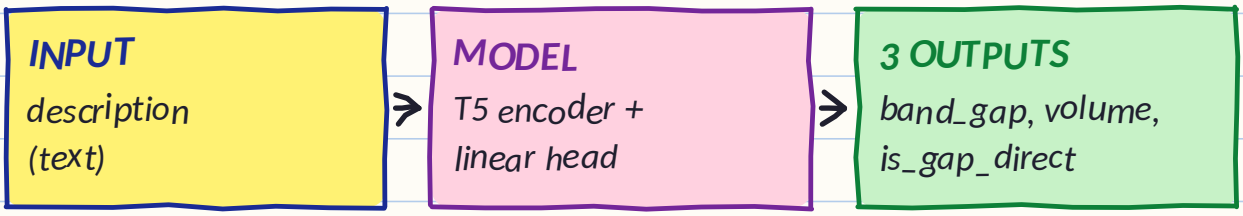
output: `nothing to commit, working tree clean`

✓ Clean = data/ is ignored, so the 700 MB can never go to GitHub by mistake.

4. The 7 columns of TextEdge

what each one is and what I use it for

`material_id`	ID of the crystal, e.g. mp-759684 (label only)
`formula`	chemical formula, e.g. Na3Bi(P2O7)2 (label only)
`cif_structure`	raw atom structure, NOT used (we use text)
`description`	crystal described in English = INPUT
`band_gap`	a number (0.0 in row 1) = OUTPUT 1
`volume`	a number (633.18 in row 1) = OUTPUT 2
`is_gap_direct`	True / False = OUTPUT 3



Task types:

- band_gap, volume = REGRESSION (predict a number, loss = MAE)
- is_gap_direct = CLASSIFICATION (yes/no, loss = BCE, score = AUC)

A sample text starts like: 'Na3Bi(P2O7)2 crystallizes in the triclinic P-1 space group. There are three inequivalent Na sites. In the first Na site, Na is bonded to five O atoms ... bond distances ranging from 2.30-2.41 A ...'

NOTE: The text has special symbols (subscripts, charges, the Angstrom sign). Block 2 will clean this.

5. First save to GitHub

steps 26 to 28

26 Refresh the library list

```
> pip freeze > requirements.txt
```

- The old list was saved before pandas etc. were in the venv, so I rebuild it.

27 Add a Day 2 note to notes.md

```
> echo.>> notes.md
> echo ## Day 2 - Dataset>> notes.md
> echo TextEdge downloaded. Files: train.csv, validation.csv, test.csv>> notes.md
> echo Columns: material_id, formula, cif_structure, description ...>> notes.md
> echo Input = description. Outputs = band_gap, volume, is_gap_direct.>> notes.md
```

- echo text >> file adds one line at the end of the file.

```
> git status
```

✓ It listed exactly 2 modified files: notes.md and requirements.txt.

28 Commit and push

```
> git add notes.md requirements.txt
> git commit -m "Day 2: download TextEdge dataset, add notes, refresh requirements"
> git push
```

TIP: Check GitHub: the new commit message appears next to notes.md and requirements.txt.

6. explore_data.py

steps 29 to 31: my first real script

29 Put the script in a src folder

```
> mkdir src
> move "%USERPROFILE%\Downloads\explore_data.py" src\explore_data.py
> dir src
```

✗ I first got: can't open file ... src\explore_data.py: No such file. Reason: the file was not in src yet. dir src shows what is really there.

30 Run it

```
> python src/explore_data.py
```

- It takes 1-2 minutes (reads 569 MB). It skips the heavy cif_structure column and prints 7 sections:

```
1 row counts 2 missing values 3 band gap
4 volume 5 direct vs indirect 6 text length
7 one sample description + saves
results\day2_distributions.png
```

31 Save to notes.md, commit, push

```
> git add notes.md requirements.txt src/explore_data.py results/day2_distributions.png
> git commit -m "Day 2: explore TextEdge data, add explore_data.py and notes"
> git push
```

✓ GitHub showed: 7407b21..ef659fb main -> main = success!

NOTE: The warning 'LF will be replaced by CRLF' is only a Windows line-ending message. It is harmless.

7. What the numbers told me

real results from the full dataset

Rows

train 125,098 | validation 9,945 | test 11,531

so: Paper says test = 9,888. My download is a slightly different version. Use my files and mention it in the report.

Missing

33 train rows have no volume / is_gap_direct

so: Drop these 33 rows in Block 2.

Band gap

48.8 percent are exactly 0 (metals). median 0.0228, max 17.89

so: Labels are very lopsided: half are zero.

Volume

median 320, mean 481, max 18,368

so: Few huge values pull the mean up. Z-score scaling helps.

Direct gap

False 107,078 (about 86 percent) | True 17,987 (about 14 percent)

so: Imbalanced, so we judge with AUC, not accuracy.

Text length

median 224 words, 29 percent are over 400 words, longest 9,825

so: Long texts get cut at the token limit (888).

REMEMBER: Viva answer: 'Why AUC and not accuracy?' Because about 86 percent of crystals are indirect-gap, so a model that always says indirect already gets 86 percent accuracy. AUC measures how well it ranks direct vs indirect.

8. Problems I hit and the fix

- ✗ (venv) not showing
  - fix: open terminal in LLM_PROP, run venv\Scripts\activate
- ✗ No module named 'pandas'
  - fix: libraries not in venv: activate it, then pip install again
- ✗ can't open file src\explore_data.py
  - fix: file is not in src: run dir src and move it in
- ✗ Google Drive virus-scan page
  - fix: click 'Download anyway'
- ✗ Excel freezes on train.csv
  - fix: never open it in Excel, use pandas
- ✗ Two nested folders TextEdge_v.1
  - fix: real path is data\TextEdge_v.1\TextEdge_v.1\

My daily routine (every day!):

1. Open a terminal inside LLM_PROP (address bar > cmd)
2. venv\Scripts\activate and check (venv) shows
3. Do the day's work
4. git add ... > git commit -m "..." > git push

9. Next: Day 3 (Block 2)

Text Preparation: clean the description text

The authors' recipe (from their code):

- Remove stopwords (the, is, of ...) using their list in stopwords/en/*.txt, but keep digits.
- Replace bond distances with [NUM]:

```
> \d+(\.\d+)?(?:–\d+(\.\d+)?)?\s*Å    ->   [NUM]
```

(the real pattern has the Angstrom sign and an en-dash)

- Replace bond angles (degree sign) with [ANG].
- Put [CLS] at the very start of each text.
- Then: drop the 33 empty rows and scale the labels (z-score).

First command of Day 3:

```
> git clone https://github.com/vertaix/LLM-Prop.git data\LLM-Prop-ref
> dir data\LLM-Prop-ref\stopwords\en /b
```

(the authors' repo is only for reading. We write our OWN code.)

Progress:

✓ Day 1 done | Day 2 done | 18 days left

Block 1 done. Keep the GitHub streak going! ✳ ✳